

Lab: Perception Detectives -- When Your Brain Tricks You

Objective

Learning Goal: Discover how your brain uses predictions to shape what you perceive, and investigate what happens when those predictions go wrong.

This is a **group challenge** designed for middle school students (Grades 6-8).

Materials Needed

- Printed optical illusions (or a device to view them)
 - Blindfold or eye mask
 - Various small objects for a touch test (coin, eraser, key, paperclip, etc.)
 - Noise-canceling headphones or earplugs (optional)
 - Lab journal or notebook
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Part 1: The Prediction Game (10 min)

Your brain does not just passively receive information -- it actively PREDICTS what it expects to see, hear, and feel. Perception is your brain checking its predictions against reality.

Try this: One partner closes their eyes. The other partner places a common object in their hand. The "blind" partner has 5 seconds of touch only to guess what it is.

Trial	Object	Prediction (First Guess)	Correct?	How confident were you (1-5)?
1				
2				
3				
4				
5				

Write your response here

Part 2: Illusion Investigation (15 min)

Look at the following types of optical illusions (your teacher will provide examples or you can search for them):

1. **Size illusion** (e.g., the Ebbinghaus illusion -- circles that look different sizes but are the same)
2. **Motion illusion** (e.g., a static image that appears to move)
3. **Color illusion** (e.g., two squares that are the same color but look different)

For each illusion, fill in this table:

Illusion	What you SEE	What is ACTUALLY there	Why does your brain get it wrong?
Size			
Motion			
Color			

Write your response here

Part 3: Social Perception Challenge (10 min)

Perception is not just about eyes and ears -- you also perceive social situations. Your brain predicts how people will act based on past experience.

Scenario sort: Read each scenario and write what you would PREDICT happens next, then what ACTUALLY might happen.

Scenario	Your Prediction	A Surprising Alternative
A classmate you do not know well smiles at you in the hallway		
Your teacher hands back a test face-down		
Your best friend has not texted you back in 3 hours		
You hear laughing from a group as you walk by		

Write your response here

Part 4: Perception Journal (5 min)

For the rest of today, try to catch your brain making predictions. Every time you notice a prediction (right or wrong), jot it down.

Time	Situation	What I predicted	What actually happened	Prediction error?

Write your response here

Reflection Table

Question	Your Answer
Is perception just recording reality? Why or why not?	
What is a prediction error?	
Why do optical illusions work?	
How do social predictions affect how you treat people?	
How does Active Inference explain perception?	

Write your response here